

Higher-Order Metaphysics

Week Ten: The place of individuals in higher order metaphysics

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1 *Materialism and beyond*

Materialism Everything is a material object.¹

¹ Alternatively: 'concrete'; 'non-abstract'; 'spatially located'; ...

This faces pressing, simple objections:

Redness/the number two/the smell of chocolate/...is not a material object.
Therefore, not everything is a material object.

A possible response: on its intended interpretation, 'everything' in the statement of Materialism is a *first-order* quantifier (a.k.a. type *e*). But 'redness', 'the number two', etc. are not semantically of type *e*, so the inference is not valid.

I think one had better agree that there is a reading of 'Not everything is a material object' on which it does follow from 'Redness is not a material object'. But the Materialist_e can say that this is a *higher-order* quantifier.²

² E.g. it might be type $e \rightarrow t$. Or it might be assigned some sum-type such as $(e + e \rightarrow t)$.

The first order of business for today is to discuss the option of taking the same attitude towards more radically minimalistic ontological hypotheses, such as those that might be inspired by various theories in physics. Examples:

Particulism Everything is an elementary particle.

Particle-or-instantism Everything is either an elementary particle or an instant of time.

Pointillism Everything is a spacetime point.

⇐ I'll initially focus on this one.

Regionalism Everything is a spacetime region.

State-space substantivalism Everything is a point/region in some high-dimensional "state space".

We could also consider the following more purely metaphysicsy doctrine:

Fundamentalism Everything is fundamental.

Such views face pressing objections:

{This chair is/that bucket is/NYU is/I am} not {an elementary particle/an instant/a point/a region/fundamental}. Therefore, not everything is {an elementary particle/an instant/a point/a region/fundamental}

But one could respond as before that ‘This chair’ (‘that bucket’, ‘Lisbon’, ‘I...’) is not semantically of type e , so that for the conclusion to follow from the premise, ‘everything’ must be interpreted as a higher-order quantifier of some sort.³

Such views have a distinguished history:

So that you would find that the logical status of Piccadilly is bound up with the logical status of series and classes, and if you are going to hold Piccadilly as real, you must hold that series of classes are real, and whatever sort of metaphysical status you assign to them, you must assign to it. As you know, I believe that series and classes are of the nature of logical fictions: therefore that thesis, if it can be maintained, will dissolve Piccadilly into a fiction. Exactly similar remarks will apply to other instances: Rumania, Twelfth Night, and Socrates.’ (B. Russell 1918–9)

³ I’ll discuss some possibilities for what the alternative type might be below.

2 *Is this even intelligible?*

Worry: the expression ‘first order quantification’ is a term of art which is anchored by paradigms like ‘some chair squeaks’, ‘every man is mortal’, etc. It is mysterious what someone could even mean by denying that these involve genuine type- e quantification.

- We could try to get a grip on the idea by trying to imagine planets where they speak variants of Higher-Orderese in which the communicative role of expressions like ‘this chair’, ‘I’, etc. is played by expressions of some appropriately higher type σ , higher than e , and the communicative role of words like ‘chair’ is played by words of type $\sigma \rightarrow t$.
- Sentences like Pointillism _{e} would not seem blatantly counterexample-prone to people like that; they may seem attractive if the physics works out appropriately. Are they doomed to error?

3 *Big things as properties of points*

In the setting of Pointillism _{e} , a natural candidate for σ —the type of names of ordinary objects like ‘Gottlob Frege’, ...—is $e \rightarrow t$.⁴ Ordinary count nouns (‘table’, ‘bucket’, ‘fish’, ‘person’) and verb phrases (‘swims’, ‘loves sardines’), correspondingly, have meanings of type $(e \rightarrow t) \rightarrow t$.⁵

⁴ Every ordinary object is a property of points.

⁵ If I’m right about pervasive type-ambiguity, such predicates will also have meanings in tons of other types, but those meanings will generally be boring (e.g. necessarily uninstantiated).

Which property of points is identical to a given ordinary object? What does a point x have to be like for it to be true that **lisbon** x , or **frege** x , or whatever?

Well, let's suppose we are already comfortable with the idea that ordinary objects are *located* at spacetime points. Then we can propose that *being located at* is just *being instantiated by*. In semantic terms, 'located at' expresses $\lambda x^e Y^{e \rightarrow t}. Yx$, of type $e \rightarrow (e \rightarrow t) \rightarrow t$.

Given β , it follows that any ordinary object is the property of being a point at which that object is located.

This view faces many objections that also apply to certain weaker views. To engage with them, it is worth pulling out some controversial consequences of the view, in ascending order of strength:

Abundance For every property F , there is an object x that is located at all and only the points that instantiate F .

Location Plenitude For every property F , there is an object x that is, necessarily, located at all and only the points that instantiate F .⁶

Identity-strength Location Plenitude For every property F , there is an object x such that to be a point at which x is located is to be a point that has F .

⁶ Dorr, Hawthorne, and Yli-Vakkuri (2021: ch. 11) argues for Location Plenitude on entirely independent grounds.

4 Objections to Abundance

Abundance already implies that, e.g., there is an object that is located exactly at those points where either my nose or the Eiffel Tower is located. Some find this absurd.

- On the other hand, a theory that only posits objects corresponding to familiar categories may seem objectionably uncharitable to mild variations of our actual practices, and unbelievably unsystematic.

Abundance also implies that there are objects located even in the intergalactic voids.

- But maybe this is fine so long as we don't read 'object' as 'material object'?

If you really wanted to block these consequences of the properties-of-points, you could consider variants where ' x is located at y ' means ' y instantiates x , and every point that instantiates x is "matter-filled"⁷, or ' y instantiates x , and the points that instantiate x are appropriately compact and demarcated from other points'.

⁷ One had better hope that physics will provide some helpful candidate to fill in this black box.

- But maybe this seems like it's kind of cheating.

5 Objections to Location Plenitude

What Location Plenitude adds to Abundance is that every object *coincides*⁸ with infinitely many others, which between them have all sorts of modal profiles. Even among friends of Abundance, some find this further multiplication incredible.⁹

But there are strong arguments that ordinary objects do, routinely, coincide. For example, many statues coincide with lumps of clay (some permanently). But no statue is *identical* to any lump of clay, since any lump of clay could, whereas no statue could, survive being squashed into a ball.

Once you have a bit of coincidence, considerations of charity (including to our own counterfactual selves), plus simplicity/systematicity, can provide alternative arguments for Location Plenitude.

Some discussions of views like Location Plenitude assume that proponents think that every *chair* coincides with many other *chairs*; every person coincides with many other people; every conscious subject coincides with many other conscious subjects; etc. That would be wild! But proponents of Location Plenitude do not have to accept it. Instead, they can say that ordinary count nouns like ‘chair’, ‘person’, and ‘conscious subject’ express properties whose instances do not (at least under ordinary circumstances) coincide, or even overlap.

- This leads to the view that something can be very similar to a chair without being a chair (etc.), at least if similarity is interpreted in such a way that physics is a guide to it. But we had better get used to saying that sort of thing, since otherwise the Sorites paradox will lead us into disaster.
- Words like ‘chair’ are plausibly quite *vague*—on a popular approach, that means that they have many “admissible precisifications”. We can say that every chair coincides with many things that fall under some precisification of ‘chair’ while also saying that each precisification of ‘chair’ is a property that is never instantiated by two coincident objects.
- From a physicalist perspective, the reasons for thinking ‘conscious subject’ is vague look similar in strength to the reasons for thinking ‘chair’ is vague.

6 Objections to Identity-Strength Location Plenitude

Note that Identity-Strength Location Plenitude follows from the combination of Location Plenitude with Intensionalism (the view that ne-

⁸ x coincides with $y := x$ is located at exactly the same points as y .

⁹ (Lewis 1986: 252) says it ‘reeks of double counting’.

cessarily coextensive properties/propositions/relations are identical). So if one wants to get off the boat at this point one will need do so in the context of some non-Intensionalist approach to questions of grain.

One objection might go like this ‘For any object x , the property of being a point at which x is located is *about* x and not about anything else. But some properties are about more than one object, or about none. So some properties F are such that there is no x such that being an F point is being a point at which x is located’.

- The required ideology of “aboutness” can’t be extended to arbitrary types: if we want to say that the proposition that *Cleopatra is a self-lover* is about self-love, β we’ll have to say that the proposition that Cleopatra loves Cleopatra is too. The claim that it works for material objects would require independent motivation.

7 Objections to the identification of objects with properties

- ‘Every chair is a property’ is a *category mistake*, and must therefore be rejected as meaningless or false.
 - Such judgments strike me as much too theoretical to carry argumentative weight. Sentences like ‘This chair is a property’ are *weird*, but ‘weird’ should not be conflated with ‘obviously false’.¹⁰
- Whatever about chairs, *I* am manifestly not a property.¹¹
 - The history of philosophy suggests to me that the metaphysical interpretation of our *introspective* knowledge is, if anything, even less clear than the metaphysical interpretation of our knowledge of ordinary material objects.

¹⁰ However, if you were really set on making ‘This chair is a property’ come out false, you could do so using the technology of sum-types from last time. ‘Chair’ and ‘property’ could have meanings of type $(et + et) \rightarrow t$, where the meaning of chair entails $\lambda x.\exists_{et}y(x = \text{inleft } y)$ while that of ‘property’ is equivalent to $\lambda x.\exists_{et}y(x = \text{inright } y)$.

¹¹ ‘I think, therefore $\exists_e x(x = \text{me})$ ’.

8 Variants

In the setting of Particulism with an A-theory of time, we can still have ordinary macroscopic objects be of type $e \rightarrow t$. In the setting of Particle-or-instantism, it’s natural to put them in type $e \rightarrow e \rightarrow t$ instead: they are relations between particles and instants. With other physics-inspired visions for what might be going on at type e —e.g. versions of state-space substantivalism—one might want still higher types. It’ll go best if one does not expect to be able to identify the right type from the armchair.

9 Eliminating individuals altogether?

Suppose we can make sense of the idea that a great many apparent objects can be “eliminated” in the sense that when we quantify over them, the quantifiers are not really type e . Could *all* objects be “eliminated” in this way, leaving only the “propositional” types $t, t \rightarrow t, t \rightarrow (t \rightarrow t), (t \rightarrow t) \rightarrow t, \dots$?

- This idea raises a new bunch of intelligibility worries. One way to try to get a grip on them is to try to imagine some aliens getting by with a version of Higher-Orderese with only the propositional types. Would they be missing out on anything? Could they formulate a true semantic theory for us?

If we managed to dispense with e -involving types, would that mean that there was some important sense in which the following is true?

Nihilism There is nothing.

Not obviously! While the aforementioned aliens would have reason to regard our quantifiers as ambiguous between many types, but it’s unclear why they’d posit a reading that makes Nihilism true.

Still, the elimination will plausibly entail that ‘Everything is either a proposition, a property, or a relation’ is true on all its readings; and given the philosophical uses of the word ‘object’ and ‘individual’, that suggests that there’s a significant true reading of ‘There are no objects’ and/or ‘There are no individuals’. I’ll refer to the view as “Nihilism” for the sake of having a label.

10 What might Nihilism look like?

Suppose we had a successful “Democratean” Pointillistic physical theory T , using a basic unary predicate ‘filled’, as well as various polyadic predicates (‘**between**’; ‘**greaterMassDensity**’; ...). Then we might hypothesize that *each spacetime point is identical to a proposition*: the proposition that it is filled.

- We rewrite T by replacing each predicate of some type $e \rightarrow \dots \rightarrow e \rightarrow t$ with one of type $t \rightarrow \dots \rightarrow t \rightarrow t$ ¹², and replacing all unrestricted type- e quantifiers with type- t quantifiers restricted to points.
- We’ll also need a predicate **point** : $t \rightarrow t$ to restrict the quantifiers. This doesn’t have to be taken as a new primitive—for example, we might consider defining it as $\lambda p. \exists q \exists r (\mathbf{between} \ pqr)$.

¹² ‘Filled’ goes to $\lambda p.p$; all the other constants, e.g. **between**, go to new constants of the appropriate type.

- Ordinary objects are now of type $t \rightarrow t$ (the same as negation!) We might propose that *being located* entails *being such that one could not be instantiated by anything that was not a point*.¹³

¹³ Or we could learn to live with an even greater multiplication of coincident objects.

11 *Making it more physically realistic*

What if physics doesn't co-operate by providing an appropriate predicate to play the role of 'filled'? We could just leave it out. But this would leave Nihilists with a weirdly weak theory, silent on the question which spacetime points are true.

- They could add a new law that says that they are all true, or that they are all false, or that all the ones where such-and-such field is nonzero are true and the others false.... But that does somewhat compromise the simplicity of the new theory relative to the original e -involving theory T , which required no corresponding law.

A perhaps more promising avenue is to look for inspiration to formulations of physical theories in terms of some kind of *state space*.

- Some authors (Albert 1996; North 2021) suggest a "substantialist" approach to such theories: points of state-space are the fundamental objects; at any time one of them (the 'marvelous point') is 'lit up' by a special fundamental property; the dynamical laws concern how the "light" moves around in response to the permanent structure of the state-space; particles and points of space are somehow derivative on this.
- This looks nicer, and less artificial, if we identify state-space points (and regions) with propositions, and "being lit up" with being true ($\lambda p.p$).

12 *Nihilism and qualitativenes*

There is a widely-used contrast between *qualitative* properties/relations/propositions (paradigms: *being red*, *something is red*) and *haecceitistic* properties (paradigm: *containing this particular electron*, *this particular point has zero mass-density*). Often, the contrast is explained in terms of individuals: to be haecceitistic is to be "about" one or more individuals, or "sensitive to" which individuals are which, or some such.

Nihilists face a terminological question here. Should they say that since there are no individuals, everything is qualitative?¹⁴ Or should they try to follow the paradigms?

¹⁴ Bacon (2019) talks this way.

There are people called *Qualitativists* who have endorsed doctrines like ‘All truths are metaphysically necessitated by the qualitative truths’.¹⁵ They have generally been deeply averse to the view that there are non-actual metaphysical possibilities where, e.g., the arrangement of particles (pattern of field-values) is the same but different particles (spacetime points) play different roles in the pattern.¹⁶ These in this camp are not going to be content with any view that retains such possibilities, whether or not they are explained in such a way that the items swapping roles are *individuals*.

It is unclear whether Nihilism does anything to undercut the considerations that lead some to think that such role-swapping possibilities are possible.

13 Nihilism and fundamentality

Another category that has been widely used in recent metaphysics is the notion of a *fundamental* (or “logically simple” or “perfectly natural”) property or relation.¹⁷

- The fundamental properties and relations are held to be in some important sense *complete*. E.g., maybe all truths *supervene* on the truths about what objects there are and which instantiate each fundamental property and relation. Or, all truths are *expressible* in some (infinite?) language with only constants for the fundamental properties and relations (plus logical constants? plus names for individuals?). Or, all truths are *necessitated* by such truths....
- The fundamental properties and relations are held to be in some important sense *independent* of one another—e.g. to obey some ‘recombination’ or ‘combinatorial’ principle.¹⁸
- Physics is held to have a key role to play in the identification of fundamental properties and relations. We seek laws that are *simple* when stated in fundamental terms.

The dominant tradition assumes (in effect) that fundamental properties and relations are all of type $e \rightarrow \dots \rightarrow e \rightarrow t$. But if Nihilists want to talk of fundamentality, they will need to relax this and take seriously that, e.g., **fundamental** ^{$(t \rightarrow t \rightarrow t \rightarrow t) \rightarrow t$} **between** ^{$t \rightarrow t \rightarrow t \rightarrow t$} .

Since propositions (type t) are just the limiting 0-adic case of relations, there is pressure even on the dominant view to allow that it makes sense to ask whether a proposition (type t) is fundamental. But one wouldn’t standardly suppose that anything of type t is fundamental—a

¹⁵ E.g. Dasgupta 2009, J. S. Russell 2018.

¹⁶ Most philosophers of physics seem to think that this is bad.

¹⁷ Dorr 2019; Dorr and Hawthorne 2013; Lewis 1983

¹⁸ Bacon 2020; Hawthorne and J. S. Russell 2018; Lewis 1986.

fundamental zero-adic property seems to do vastly less useful theoretical work than, e.g., a fundamental binary relation.¹⁹

But there is some pressure on Nihilists to think that *lots* of propositions are fundamental—e.g. all the spacetime points/statespace points. For example, it is hard to see how they could all be *definable* in terms of some short list of fundamental higher-type entities.²⁰

Or maybe there is a big infinite stock of fundamental things in some higher propositional type: for example, Bacon (2019) describes a view each particle (in the old theory) corresponds to a new fundamental operator of type $t \rightarrow t$.

If this sort of multiplication is inevitable for the Nihilist, it suggests some objections.

- An objection from *economy*. By admitting type e , we only have to posit a *few* fundamental entities (properties and relations among individuals). By contrast, in eliminating type e we have had to posit a large infinite number of fundamental entities.
 - But it is not obvious that this is the right way to think of what’s happened. One might think that even if they didn’t bother with a **fundamental** predicate of type $e \rightarrow t$, the non-Nihilists were in effect treating all individuals as fundamental. The Nihilist just switches the type in a way that doesn’t make any obvious difference from the point of view of economy.
- An objection from *combinatorialism*. If one accepts that the only sentences in a fundamental language that express necessary truths are those that are logical truths (in some sense), and thinks that such a language would have a type- t constant for each point, one will be led to some very strange modal judgments.

For example, we’ll have to say it’s possible for all state-space points to be “lit up” (true) simultaneously, if they are fundamental. And if we take **between** to be fundamental, we’ll have to say it’s possible that **between** $\top\top\top$, that $\neg$ **between** $\top\top\top$, etc.

- Note however that even in the familiar setting where we do have individuals, combinatorialism leads to some quite strange modal judgments!²¹ Against this background, appeals to “modal intuitions” don’t have as much force.

¹⁹ There are other meanings of ‘fundamental’ floating around that seem to *only* take propositional arguments, e.g. ones explained in terms of ungroundedness. Don’t conflate these with the one that’s paradigmatically about properties.

²⁰ Indeed, on the “filled” version and the “marvelous point” version, they don’t even *nominally* supervene on the truths about betweenness etc.

²¹ E.g., if **between** : $e \rightarrow e \rightarrow e \rightarrow t$ is fundamental, we’ll have to say that $\Diamond \exists_e x \exists_e y (\text{between } xxx \wedge \neg \text{between } yyy)$.

14 Nihilism and the complexity of the laws

Even apart from the issue of whether the points themselves are fundamental, there are also objections that arise because of the fundament-

ality of higher-type predicates. For example, suppose we want to say that betweenness is a fundamental relation. In the old theory T where **between** was of type $e \rightarrow e \rightarrow e \rightarrow t$, the (geometrical/physical) laws only needed to tell us about how betweenness behaves on the points. In the Nihilist setting, we'll also need laws to tell us how betweenness behaves on propositions that aren't points.

These laws could be *fairly* simple—e.g. we could just say that whenever **between** pqr , p , q , and r are all points, so that, e.g.

$$\forall pqr. \text{between } pqr \rightarrow \neg \text{between}(\neg p)q'r'$$

Or we could say that **between** pqr is true whenever there are points p', q', r' such that **between** $p'q'r'$, p' entails p , q' entails q , and r' entails r . There are many possibilities.

Still, these laws and the need to choose between them don't correspond to anything in the old theory.

References

- Albert, David Z. (1996), 'Elementary Quantum Metaphysics', in James T. Cushing, Arthur Fine, and Sheldon Goldstein (eds.), *Bohmian Mechanics and Quantum Theory: An Appraisal* (Dordrecht: Kluwer), 277–84.
- Bacon, Andrew (2019), 'Is Reality Fundamentally Qualitative?', *Philosophical Studies*, 176: 259–95.
- (2020), 'Logical Combinatorialism', *The Philosophical Review*, 129/4: 537–89.
- Dasgupta, Shamik (2009), 'Individuals: An Essay in Revisionary Metaphysics', *Philosophical Studies*, 145/1 (Apr.): 35–67.
- Dorr, Cian (2019), 'Natural Properties', *Stanford Encyclopedia of Philosophy*, Fall 2019, ed. Edward N. Zalta, <https://plato.stanford.edu/entries/natural-properties/>.
- Dorr, Cian, and Hawthorne, John (2013), 'Naturalness', in Karen Bennett and Dean Zimmerman (eds.), *Oxford Studies in Metaphysics*, viii (Oxford: Oxford University Press), 3–77.
- Dorr, Cian, Hawthorne, John, and Yli-Vakkuri, Juhani (2021), *The Bounds of Possibility: Puzzles of Modal Variation* (Oxford: Oxford University Press).
- Hawthorne, John, and Russell, Jeffrey Sanford (2018), 'Possible Patterns', in Karen Bennett and Dean W. Zimmerman (eds.), *Oxford Studies in Metaphysics*, xi (Oxford: Oxford University Press), 149–92 (forthcoming).
- Lewis, David K. (1983), 'New Work for a Theory of Universals', in id., *Papers in Metaphysics and Epistemology* (Cambridge: Cambridge University Press, 1999), 8–55. From *Australasian Journal of Philosophy*, 61 (1983): 343–77.
- (1986), *On the Plurality of Worlds* (Oxford: Blackwell).
- North, Jill (2021), *Physics, Structure, and Reality* (Oxford: Oxford University Press).
- Russell, Bertrand (1918–9), *The Philosophy of Logical Atomism*, ed. David F. Pears (La Salle, Illinois: Open Court, 1985). From *The Monist*, 28 (1918): 495–527; 29 (1919): 32–63, 190–222, 345–380.
- Russell, Jeffrey Sanford (2018), 'Quality and Quantifiers', *Australasian Journal of Philosophy*, 96/3: 562–77.